

Junyi (Conny) Zhou

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EDUCATION

- Harvard University** Boston, MA
M.S. in Health Data Science *August 2025 – December 2026*
 - Coursework: Deep Learning, Healthcare Machine Learning
- Emory University, College of Arts and Sciences** Atlanta, GA
B.S. in Applied Mathematics and Statistics; Minor in Computer Informatics; GPA: 3.925 *August 2021 – May 2025*
 - Honors: Dean's List Spring 2022, Spring 2023
 - Coursework: Machine Learning, NLP, Probability and Statistics, Data Structures, Database Systems

TECHNICAL SKILLS

- Languages:** Python, SQL, C++, Java, R, JavaScript
- ML / AI:** PyTorch, TensorFlow, Keras, Hugging Face Transformers, scikit-learn, XGBoost, LightGBM
- LLM Applications:** RAG, agentic pipelines, prompt engineering, context-window management, OpenAI APIs, Pinecone
- Data & Cloud:** PostgreSQL, MySQL, MongoDB, Spark, Hadoop, AWS (SageMaker, Lambda, S3, EC2), Snowflake, BigQuery
- Certifications:** AWS Machine Learning Associate; AWS AI Practitioner; AWS Cloud Practitioner

SELECTED EXPERIENCE

- Agentic LLM + RAG Simulation System** Atlanta, GA
Full-Stack Engineer & Scrum Master — Emory Pediatric Hospital *January 2024 – Present*
 - Built an **agentic LLM pipeline** with **RAG** and OpenAI models to ground multi-turn responses and reduce hallucinations under privacy constraints.
 - Implemented **prompt engineering and context-window management** for dynamic scenario generation and stateful practice sessions.
 - Shipped analytics-backed product stack (**Python/Flask**, React, MongoDB, AWS) used to track learner performance in real time.
 - Led a team of 6; iterated NLP/LLM behavior from end-user feedback. Demo: CloudFront.
- Transformer Architecture Reimplementation** Atlanta, GA
Machine Learning Researcher — Emory Department of Computer Science *October 2024 – January 2025*
 - Reimplemented a Transformer in **PyTorch** (multi-head attention, positional encodings, layer norm) and validated against `nn.Transformer`.
 - Optimized training and hyperparameters, achieving **9.38 BLEU** on English-to-German translation.
 - Code: GitHub.
- Generative Models & Optimal Transport** Boston, MA
Graduate Researcher — Mooney Lab & Alvarez-Melis Lab, Wyss Institute *February 2026 – Present*
 - Building **PyTorch** frameworks with VAEs, diffusion models, and Optimal Transport / CellOT for out-of-distribution mapping in high-dimensional data.
 - Designing scanpy-based preprocessing and evaluation pipelines over large public and lab-generated datasets.
- Large-Model Inference Pipelines on HPC** Atlanta, GA
Data Analyst — Bou-Nader Lab, Emory Department of Biochemistry *February 2025 – August 2025*
 - Engineered high-throughput inference workflows (AlphaFold-Multimer / AlphaFold3) with GPU parallel scheduling on HPC.
 - Built scoring/filter scripts (pLDDT, PAE) and CLI tooling for non-ML researchers. GitHub.